

Shammika Abhiman Wijesinghe

Computer Engineer

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INTRODUCTION

I am a final-year undergraduate studying Computer engineering at the Computer Engineering Department, University of Ruhuna. My goal is to enhance my knowledge and skills in DevOps, Cloud technologies, AI, DevSecOps, and other sciences and apply them to push the boundaries of computer engineering.

EDUCATION

Faculty of Engineering, University of Ruhuna	<i>March 2022 – present</i>
Bachelor of Science (Hons) in Engineering - Specializing in Computer Engineering	<i>SGPA : 3.5/4.0</i>
School of Computing, National Institute of Business Management	<i>March 2021 - March 2022</i>
Diploma in Software Engineering	<i>OGPA : 3.93/4.0</i>
Dharmaraja College Kandy	<i>Jan. 2012 - May. 2021</i>
GCE Ordinary Level - 2017	<i>9As</i>
GCE Advanced Level (Physical Science Stream) - 2020	<i>2As 1B</i>

PROJECTS AND WORK EXPERIENCES

GTN Technologies – Intern DevOps Engineer *May 2025 - Nov 2025*

- **Technologies:** AWS (EC2, ECS, ECR, S3, Batch, EventBridge, SQS, Lambda, AppConfig), Linux, Terraform, Ansible, GitLab CI/CD, Git, Docker, Datadog, Grafana, Loki, Alloy, Fluent Bit.
- **Key Contributions:**
 - Designed and implemented an event-driven architecture using AWS EventBridge, SQS, and Lambda for the Asset Manager Terminal Back-office.
 - Developed and refactored AWS Batch infrastructure from scratch using Terraform modules across Dev, QA, Demo, and Prod environments.
 - Implemented centralized logging pipelines (ECS/Batch to Datadog & Grafana) with environment-based switching (Loki/Datadog).
 - Created and configured Datadog monitors (CPU, DB, error alerts) including agent installation and production alerting.
 - Enhanced GitLab CI/CD pipelines (multi-pipeline triggering, scheduler infrastructure separation, automation fixes).
 - Built ECR modules and automated container workflows for job schedulers.
 - Implemented configuration management using AWS AppConfig and S3.

Cyber Threat Intelligence (CTI) Platform for Open RAN *Ongoing*

- **Functionality:** Designed and implemented a disaggregated Cyber Threat Intelligence (CTI) platform native to Open RAN, integrating with the RAN Intelligent Controller (RIC) via custom rApps and xApps to automate threat ingestion, O-RAN-specific semantic filtering, AI-driven severity classification, RAG-based mitigation generation, and prioritized inter-platform threat intelligence sharing.
- **Technologies:** Python, FastAPI, PostgreSQL, ChromaDB, Docker, Kubernetes (GKE), Azure, GCP, srsRAN, Open5GS, O-RAN SC Near-RT/Non-RT RIC, STIX/TAXII, Sentence-Transformers, Gemini API
- **Contribution:** Developed the RAG-based mitigation generation engine using ChromaDB and LLM reasoning to map threats to affected O-RAN components with actionable, contextualized defense recommendations. Implemented xApp/rApp components for cyber probe management, threat simulation, and A1-interface policy delivery, and deployed the full testbed across Azure and Google Kubernetes Engine.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java, Dart
- **Databases:** MySQL, PostgreSQL, MongoDB, Firebase
- **Machine Learning and AI:** OpenAI, LLMs, Google Colab, Jupyter notebook, Numpy, Pandas, Seaborn
- **DevOps:** Azure, GCP, AWS, Docker, Jenkins, Terraform, Ansible, GitLab, GitHub, GitHub Actions, Linux, Kubernetes, Datadog, Grafana, Loki, Alloy, Fluentbit

LICENSES AND CERTIFICATIONS

- **Microsoft Certified: Azure Administrator Associate** *Reading*
- **AWS Certified: Solution Architect Associate** *Reading*
- **AWS Agent AI Demonstrated** - AWS micro-credentials *Credentials* [🔗](#)
- **DevOps Essentials** - IBM *Credentials* [🔗](#)
- **Docker for Developers** - LinkedIn Learning *Credentials* [🔗](#)
- **DevOps Foundations: DevSecOps** - LinkedIn Learning *Credentials* [🔗](#)
- **Manage Kubernetes in Google Cloud** - Google *Credentials* [🔗](#)
- **Learning Kubernetes** - LinkedIn Learning
- **AWS Educate Introduction to Cloud 101** - AWS
- **AWS Security Fundamentals** - AWS

ACHIEVEMENTS - ACADEMICS

- **T.Fonseka memorial prize** at the prize giving of Dharmaraja College Kandy for the student who performed well in academics and extracurricular activities for the year 2020.
- **Winner at CryptX1.0** Hackathon organized by the Faculty of Technology in University of Sri Jayawardenapura for Taprobana: AI powered travel assistant.
- **Winner of InsurgeX_1.0 Hackthon** organized by the Computer Science Students' Community of Faculty of Science University of Ruhuna.

ACHIEVEMENTS - SPORTS

- **Dagoba for Athletics**, the highest accolade of a sportsman at Dharmaraja College *2019*
could achieve at the College Colours Nite.
- **Vice Captain** of the athletic team in Engineering Faculty *2024*
- **Member** of the University Athletic and Cricket teams *2023 - 2024*

EXPERIENCES IN CLUBS AND SOCIETIES

- **President** - Sports Council of Dharmaraja College *2019 - 2020*
- **Vice President** - Sports Council of Dharmaraja College *2018 - 2019*
- **Secretary** - Aviation Club of Dharmaraja College *2019 - 2020*
- **Vice Secretary** - Astronomical Society of Dharmaraja College *2019 - 2020*

SOFT SKILLS

- **Teamwork and Team Management**
Member of college general knowledge, aviation and astro quiz teams. Member of the athletic team and the cricket team in the college and the University.
- **Leadership and Communication Skills**
Games captain in the college, Athletic Vice Captain in the University. Led the Logistics Team in the CareerXpo1.0 and led the Program Team in INSL Southern province competition.
- **Critical Thinking and Problem Solving**
Demonstrated strong analytical and problem-solving skills while participating and winning hackathon competitions, tackling complex challenges under time constraints in Application development and Software development

VOLUNTEERING

- **Logistic Team Lead** of the **CareerXpo1.0** organized by EIES, MMES IEEE *2024*
student branch of University of Ruhuna
- **Program Team Lead** of the **INSL Southern Province competition**, organized by IEEE Young Professionals Sri Lanka *2024*

REFERENCES

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| ◦ Dr. Kushan Sudheera
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DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and I take full responsibility for its accuracy.